

CyClaw Local Manual Sandbox — macOS Verification Checklist

PRs #911 (Keychain + Telegram LaunchAgents) and #912 (gate.py / harness supervised LaunchAgent)
Assumes all currently open PRs are merged into main first. Real Apple Silicon hardware required.

1. Executive Summary & Scope

PR #911 (merged 2026-08-15) shipped macos/cyclaw-keychain-*.sh and Telegram poll-plist / health-plist generators. PR #912 (open draft) adds the highest-risk generator: macos/generate_service_plist.py for supervised, KeepAlive LaunchAgents on gate.py (:8787) and harness.server (:8790). Both were developed and unit-tested only inside a Linux sandbox. All live launchctl, Keychain ACL, crash-restart, and clean-stop behavior is unverified on real macOS. This document is the single authoritative operator checklist for that verification.

Risk framing (from PR #912 body and MACOS_LAUNCHD_INTEGRATION_PLAN.md): Once loaded, the gate/harness agent makes a loopback network listener persistent across logout, reboot, and process crashes. That is a qualitative change in availability and attack surface compared with every prior generator (sync, trash, Telegram poll). Do not treat a green Linux CI run as sufficient.

2. Prerequisites (before any command)

- macOS 13+ (Sonoma preferred; torch wheel floor is macosx_14_0_arm64).
- Apple Silicon (M-series). Intel Macs unsupported for current torch pin.
- Python 3.12.x (exact contract enforced by installer).
- Fresh clone of main after merging all open PRs (especially #912). Work on a clean branch or detached HEAD.
- No pre-existing CyClaw LaunchAgents loaded. Run `launchctl list | grep cgfixit` and bootout any hits first.
- Dependencies installed: `pip install -r requirements.txt -c constraints.txt --ignore-installed PyYAML` (or use `macos/install-cyclaw.sh`).
- Terminal / iTerm has Files and Folders access if needed; no TCC automation is provided.
- You understand that loading a supervised agent is irreversible without an explicit bootout + plist removal.

3. What Must Be Verified (Summary Matrix)

Items below are the exact gaps called out in the PR bodies, the consolidated comment on #912, and docs/work/MACOS_LAUNCHD_INTEGRATION_PLAN.md. CI cannot exercise any of them.

ID	Component	What to prove on real hardware	Source
K1	Keychain set (bare -w)	Secret never appears in argv / ps; TTY-only; ACL pinned to /usr/bin/security	#911 + Codex P2 fix
K2	Keychain env wrapper	Fail-closed on missing/empty item; correct export + exec; no prompt on subsequent fetch after ACL grant	#911
T1	Telegram poll-plist	ProgramArguments chain Keychain wrappers; zero plaintext token in plist	#911
T2	Telegram health-plist	Same + single wrapper; health notify path works under launchd	#911
G1	Gate plist structure	Label, RunAtLoad=true, KeepAlive={SuccessfulExit:false}, ThrottleInterval=30, WorkingDirectory=repo	#912
G2	Gate clean stop	launchctl stop / bootout → process exits 0 and is NOT restarted	#912 (uvicorn SIGTERM)
G3	Gate crash restart	kill -9 → restart after ≥ ThrottleInterval; no tight loop	#912
G4	Gate port conflict	Manual gate already bound → agent exits 0 (pre-check) and does NOT crash-loop	#912
H1	Harness plist	Label + CYCLAW_HOME/CYCLAW_REPO env; same KeepAlive semantics	#912

ID	Component	What to prove on real hardware	Source
H2	Harness clean stop	Same as G2	#912
H3	Harness port conflict	Port held → crash-loop (no pre-check) throttled by ThrottleInterval	#912
U1	Uninstall symmetry	uninstall-cyclaw.sh boots out + removes all landed labels including gate/harness if present	#919 + #914 + #912
E1	Confirm/reason gates	Missing --confirm or blank --reason → refuse write, exit 1, no plist created	#912

4. Official Documentation Basis (verified against)

All KeepAlive / SuccessfulExit claims rest on Apple launchd.plist semantics and widely-cited community confirmation of the same contract:

- **KeepAlive dictionary with SuccessfulExit:false** — job is restarted only when the previous exit status is non-zero. A clean exit (0) leaves the job stopped. Confirmed in launchd property-list key documentation and multiple independent sources (Apple Stack Exchange launchd tutorial, tjluoma/launchd-keepalive examples, Launchyard launchd-plist-spec).
- **ThrottleInterval** — minimum seconds between successive launches of a KeepAlive job. Default launchd value is 10 s; the generator defaults to 30 s because gate.py embedding + ChromaDB startup is slower than a poller.
- **security add-generic-password -w** (bare) — when -w is supplied without an attached value it is documented to prompt for the password (secure TTY read). Putting the value on the command line is the insecure path the PR deliberately avoided. Sources: security(1) / add-generic-password usage notes (ss64, DSSW, Apple man pages).
- **-T /usr/bin/security** — pins the ACL so only the security binary (not arbitrary processes) may access the item without further prompting.

5. Detailed Step-by-Step Verification Procedures

Execute in order. Record pass/fail and any unexpected dialogs or logs. Do not skip the clean-stop tests — they are the only proof that `KeepAlive:{SuccessfulExit:false}` behaves as designed with real `uvicorn`.

Phase 1 — Keychain Wrapper Security (PR #911)

1a. Bare -w interactive store (secret never in argv)

Command: `bash macos/cyclaw-keychain-set.sh com.cgfixit.test-token-123` Expected: interactive no-echo prompt from security itself. After entry, item exists. While the security process runs, inspect with `ps auxww | grep security` (or Activity Monitor) — the secret must not appear in any process arguments. Exit 0.

1b. Non-interactive / non-TTY refusal

Command: `echo 'test-secret' | bash macos/cyclaw-keychain-set.sh com.cgfixit.test-pipe` Expected: exit code 1, stderr mentions TTY requirement. No item created under that service name.

1c / 1d. ACL pinning and subsequent fetch

After 1a, open Keychain Access.app or run `security dump-keychain -d` (or `find-generic-password -g`) and confirm the Access Control List grants to `/usr/bin/security` (not `/bin/bash` or the shell). Then exercise the env wrapper: `bash macos/cyclaw-keychain-env.sh com.cgfixit.test-token-123 TEST_TOKEN --env | grep TEST_TOKEN` Expected: value printed, no further Keychain prompt after the first ACL grant. Missing or empty item must refuse to exec (fail-closed).

Phase 2 — Telegram Plist Generation (PR #911)

Note: after #911 merge the generators live under `python -m telegram.cli poll-plist /health-plist`. Do not use the #912 service generator for Telegram.

2a. Poll plist Keychain chain

Command (after configuring `telegram.enabled + allowed_chat_ids + bot_token_env` if needed): `python -m telegram.cli poll-plist` Inspect: defaults read `~/Library/LaunchAgents/com.cgfixit.cyclaw.telegram-poll.plist` ProgramArguments or plutil -p Expected: first argument is the keychain-env.sh path, service name, env var, '--', then the real `python -m telegram ...`. No `TELEGRAM_BOT_TOKEN` string appears as a literal value anywhere in the plist.

2b. Health plist

Same inspection for `com.cgfixit.cyclaw.telegram-health.plist`. Single wrapper for the bot token. Optional live test: bootstrap, force a /health failure, confirm notify path fires (requires real bot token + allowlisted chat).

Phase 3 — Gate LaunchAgent (PR #912) — highest risk

3a. Plist structure generation

Command: `python macos/generate_service_plist.py --service gate --reason "manual verification of crash-only KeepAlive" --confirm` Expected: writes `~/Library/LaunchAgents/com.cgfixit.cyclaw.gate.plist`, prints bootstrap command, does NOT load it. Verify with `plutil/defaults`: `Label=com.cgfixit.cyclaw.gate`, `RunAtLoad=true`, `KeepAlive={SuccessfulExit=false}`, `ThrottleInterval=30`, ProgramArguments ends with `gate.py`, `WorkingDirectory = repo root`. No `--config` flag is present (gate always loads `repo config.yaml`).

3b. Port-conflict clean exit (gate pre-check)

1. Start gate manually in a terminal: `python gate.py` (or via cyclaw). 2. Bootstrap the agent: run the printed `launchctl bootstrap gui/$(id -u) ...` command. 3. Observe: agent process should exit 0 almost immediately (gate's own port-in-use check). `KeepAlive` must NOT restart it in a loop. 4. Stop the manual instance, then bootstrap again or kickstart; agent should now stay up.

3c. Clean stop — the critical KeepAlive test

With the agent running and healthy: `launchctl stop com.cgfixit.cyclaw.gate` # or `bootout` Expected: process terminates with exit 0. `launchctl list /` print should show it is not running and is not immediately relaunched. If it restarts, the `SuccessfulExit:false` contract is broken for this binary (`uvicorn SIGTERM` path). This is the single most important verification in the entire checklist.

3d. Crash-loop throttle

Temporarily force a crash (e.g. insert an early `sys.exit(1)` or kill -9 the PID while the agent is loaded). Observe restart timing in the log file (`~/Library/Logs/CyClaw/gate.log` or the path printed). Restarts must be spaced at least `ThrottleInterval` (default 30 s) apart. Restore original code and

bootout when finished.

3e. Config path honesty

Passing a non-default `--config` only affects the port the generator reports. `gate.py` always loads the `repo-root config.yaml`. Confirm the warning is printed when `--config` differs, and that the running process binds the port from the real `config.yaml`.

Phase 4 — Harness LaunchAgent (PR #912)

4a. Structure

Command: `python macos/generate_service_plist.py --service harness --reason "manual verification" --confirm Verify Label=com.cgfixit.cyclaw.harness, EnvironmentVariables contains CYCLAW_HOME and CYCLAW_REPO, same KeepAlive/Throttle semantics, ProgramArguments uses -m harness.server.`

4b. Clean stop

Identical procedure to 3c. `launchctl stop` must leave the process stopped.

4c. Port-conflict crash-loop (documented limitation)

Hold :8790 with any listener (e.g. `python -m http.server 8790` or another harness). Bootstrap the harness agent. Expected: crash-loop at `ThrottleInterval` because `harness/server.py` has no pre-flight port check. This is the documented gap, not a bug. Confirm it is throttled, not spinning. Cleanup: free the port, bootout the agent.

Phase 5 — Install / Uninstall Symmetry

After any agents have been loaded, run `bash macos/uninstall-cyclaw.sh` (add flags as needed). Confirm: `launchctl list` no longer shows `com.cgfixit.cyclaw.gate` / `.harness` / `telegram-*` / `fsconnect-trash`; plist files under `~/Library/LaunchAgents` for those labels are gone. Current main (post-#919) boots out by label even if the plist was already deleted – verify that path as well.

Phase 6 — Edge Cases & Hardening

- Missing `--confirm` → exit 1, risk text on stderr, no plist written.
- Missing or blank `--reason` → same.
- Non-Darwin platform → exit 3 (script refuses).
- Missing `config.yaml` when generating gate → exit 3.
- Optional `--api-key-service` correctly inserts the `keychain-env.sh` wrapper for both gate and harness; no 'no effect' warning for harness.

6. Sign-off Checklist (copy into PR or issue)

#	Item	Pass?	Notes / date
1	K1–K2 Keychain bare -w + ACL + fail-closed		
2	T1–T2 Telegram plists contain zero plaintext tokens		
3	G1 Gate plist structure matches contract		
4	G2 Gate clean stop does NOT restart (critical)		
5	G3 Gate crash restart respects <code>ThrottleInterval</code>		
6	G4 Gate port conflict exits cleanly (no loop)		
7	H1–H2 Harness structure + clean stop		
8	H3 Harness port conflict is throttled crash-loop		
9	U1 Uninstall removes all landed agents by label		
10	E1 Confirm/reason gates refuse write		
11	No unexpected Keychain / TCC dialogs after first grant		
12	Logs written to expected paths; no secret leakage in logs		

Operator signature / date: _____ Hardware: _____ macOS version: _____

7. Residual Risks (document, do not 'fix' away)

- Keychain 'Allow / Always Allow' dialog attribution to /usr/bin/security is expected per Security framework model but should be observed once on real hardware.
- uvicorn / gate.py / harness clean SIGTERM exit is verified by code inspection only; the clean-stop test above is the live proof.
- No TCC entitlements are requested; future launchd jobs that touch protected paths may surface additional prompts.
- Signed / notarized distribution is out of scope; this is operator-driven personal installation only.
- harness port-conflict crash-loop is intentional documentation of a missing pre-check; do not treat a tight loop as a regression if ThrottleInterval is honored.
- Once a supervised agent is loaded it is a standing loopback listener. Threat model still assumes single-operator, loopback-only, no non-loopback bind.

Primary sources: PR #911 body + Codex P2 resolutions, PR #912 body + consolidated comment, docs/work/MACOS_LAUNCHD_INTEGRATION_PLAN.md, launchd.plist KeepAlive/SuccessfulExit semantics, security(1) add-generic-password -w interactive prompt notes. Generated for operator use on real Apple Silicon after all open PRs are merged.